

Functional & Technical Design Document (FTDD)

Status: DRAFT | UNDER REVIEW | APPROVED

Version: 1.0

Date: [Insert Date]

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Repository: [Insert GitHub Repo Link]



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0.1 Versioning & Change History

Version Date Author Description

0.1 [Insert] [Insert] Initial release

Version	Date	Author	Description
1.0	[Insert]	[Insert]	Initial release

1.0 Business Context Summary

Problem Statement: [Describe the business problem or opportunity.]

Business Drivers:

- [Driver 1]
- [Driver 2]
- [Driver 3]

Stakeholders:

- [Business Owner]
- [Technical Owner]
- [End Users]

Success Metrics / KPIs:

- [Metric 1]
- [Metric 2]

2.0 Document Purpose

[Describe the purpose of this document and the audience it serves.]

3.0 Scope, Assumptions, and Constraints

3.1 In-Scope

- [Requirement 1]
- [Requirement 2]
- [Requirement 3]

3.2 Out-of-Scope

- [Excluded feature]
- [Excluded integration]

3.3 Assumptions, Constraints, and Risks

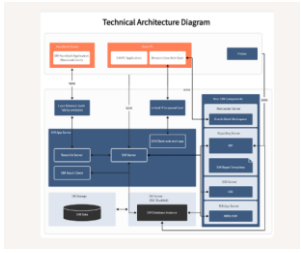
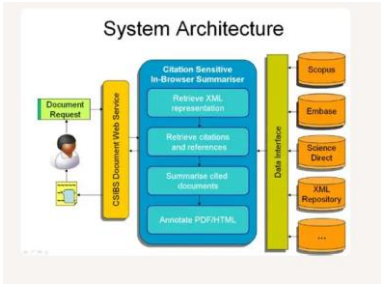
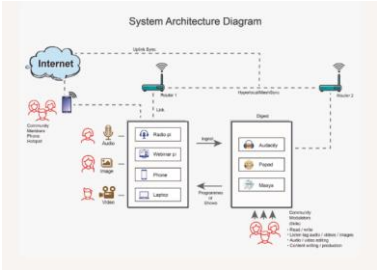
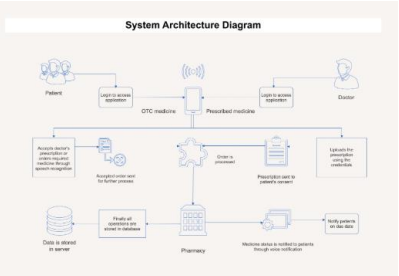
Assumptions: [e.g., API availability] **Constraints:** [e.g., regulatory limits] **Risks:** [e.g., dependency delays]

4.0 Conceptual Design

[High-level narrative of how the system solves the business problem.]

4.1 High-Level Architecture Diagram

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-
-
-



5.0 System Process Flow Design

5.1 Sequence or Swimlane Diagram

[Insert diagram or ASCII representation]

5.2 Functional Step-by-Step Logic

- **Trigger:**
- **Validation:**
- **Execution:**
- **Output:**

6.0 Non-Functional Requirements (NFRs)

6.1 Performance & Scalability

- Latency: [e.g., 200ms @ P95]
- Throughput: [e.g., 500 TPS]
- Data growth: [e.g., 100GB/year]

6.2 Availability & Reliability

- Uptime: [e.g., 99.9%]
- RTO: [e.g., 4 hours]
- RPO: [e.g., 1 hour]

6.3 Security & Compliance

- MFA + RBAC
- PII masking
- Immutable audit logs

6.4 Maintainability & Extensibility

- Automated linting
- Modular architecture

7.0 Detailed Design

7.1 Extract / Ingestion Requirements

- Sources
- Methods
- Rate limits

7.2 Software & Application Design

- Tech stack
- Design patterns
- Component breakdown

7.3 Infrastructure & Network Design

- Hosting
- Network topology
- Scaling strategy

7.3.1 Quality Environment for Testing

- Environment mirroring
- Test data provisioning

7.4 Database & Data Storage Design

- Storage engine
- Schema
- Indexing strategy

7.5 Data Conversion & Migration Design

- Migration strategy
- Rollback plan

7.6 Interfaces & API Design

- Protocols
- Payload formats

7.7 Security, Compliance, & Privacy Design

- Authentication
- Encryption
- Compliance mapping

7.8 Batch Jobs, Automation, & Schedulers

- Job definitions
- Schedules

7.9 Error Handling & Logging

- Exception strategy
- Logging standards
- Correlation IDs

7.10 Dependencies & Construction Sequence

- Prerequisites
- Phase 1
- Phase 2
- Phase 3

8.0 Supportability & Maintainability

- Health probes
- Telemetry
- Metrics

9.0 Testing & Validation Strategy

- Unit testing
- Integration testing
- Acceptance criteria

10.0 Requirements Traceability Matrix (RTM)

Req ID	Description	Design Section	Test Case
REQ-001	[Description]	Section 7.X	[Test ID]
REQ-002	[Description]	Section 7.X	[Test ID]

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REQ-001	[Description]	Section 7.X	[Test ID]
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11.0 Glossary & Definitions

Term	Definition
API	Application Programming Interface
RBAC	Role-Based Access Control
PII	Personally Identifiable Information

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